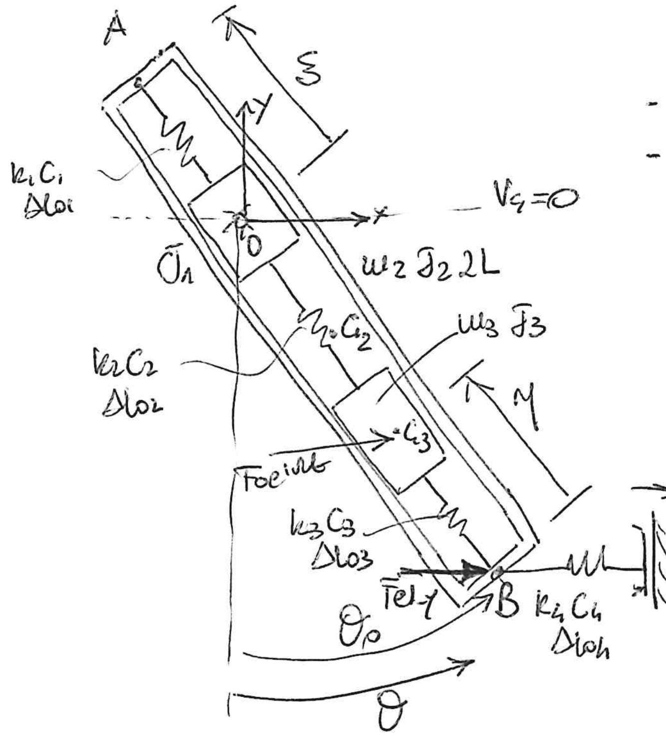
WRITTEN TEST - November 05th 2019



- 3 rigid bodies → 9 dof without constraints
- 1 hinge O + 2 slider between J_1 and m_2 and m_3 and m_2 → 6 doc → 3 dof

$$\underline{x} = \begin{Bmatrix} \theta \\ \xi \\ \eta \end{Bmatrix} \qquad \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ L_1 \\ L_2 \end{Bmatrix}$$

$$F_{el-y} = k_4 y + c_4 \dot{y}$$

$$\begin{cases} T = \frac{1}{2}(J_1 + J_2 + J_3)\dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G2}^2 + \frac{1}{2} m_2 \dot{y}_{G2}^2 + \frac{1}{2} m_3 \dot{x}_{G3}^2 + \frac{1}{2} m_3 \dot{y}_{G3}^2 \\ D = \frac{1}{2}\sum_{i=1}^{4} c_i \Delta \dot{l}_i^2 \ ; \ V_{el} = \frac{1}{2}\sum_{i=1}^{4} k_i \Delta l_i^2 \ ; \ V_g = \sum_{i=1}^{3} m_i g h_i \\ \delta W = F_0\, \delta x_{G3} + F_{el-y}\, \delta x_B \end{cases}$$

$$m_{PH} = \mathrm{diag}(J_1 + J_2 + J_3 \quad m_2 \quad m_2 \quad m_3 \quad m_3)$$

$$C_{PH} = \mathrm{diag}(C_1\ C_2\ C_3\ C_4) \qquad k_{PH} = \mathrm{diag}(k_1\ k_2\ k_3\ k_4) \qquad \underline{F} = \begin{Bmatrix} F_0 \\ F_{el-y} \end{Bmatrix}$$

$$\underline{Y}_M = \{\theta \quad x_{G2} \quad Y_{G2} \quad x_{G3} \quad Y_{G3}\}^T \qquad \underline{Y}_c = \underline{Y}_k = \{\Delta l_1\ \Delta l_2\ \Delta l_3\ \Delta l_4\}^T$$

$$\underline{Y}_Q = \{x_{G3} \quad x_B\}^T$$

- Non linear kinematic relationships:

$$x_{G2} = (L - \xi)\, \mathrm{sen}\,\theta$$

$$Y_{G2} = -(L - \xi)\cos\theta$$

$$x_B = (2L - \xi)\, \mathrm{sen}\,\theta$$

$$Y_B = -(2L - \xi)\cos\theta$$

$$x_{G3} = x_B + x_{BG3} = (2L - \xi)\,\mathrm{sen}\,\theta - \eta\, \mathrm{sen}\,\theta$$

$$Y_{G3} = Y_B + Y_{BG3} = -(2L - \xi)\cos\theta + \eta \cos\theta$$

$$\Delta l_1 = \xi - L_1$$

$$\Delta l_3 = \eta - L_2$$

$$\Delta l_2 = (2L - \xi - \eta) - (2L - L_1 - L_2)$$

$$\Delta l_4 = -\Delta x_B = -(2L - \xi)\,\mathrm{sen}\,\theta$$

$$\frac{\partial x_{s2}}{\partial \vartheta} = (L-\xi)\cos\vartheta \,; \quad \frac{\partial x_{s2}}{\partial \xi} = -\text{sen}\,\vartheta$$

$$\frac{\partial y_{s2}}{\partial \vartheta} = (L-\xi)\,\text{sen}\,\vartheta \,; \quad \frac{\partial y_{s2}}{\partial \xi} = \cos\vartheta$$

$$\frac{\partial x_{s3}}{\partial \vartheta} = (2L-\xi)\cos\vartheta - \eta\cos\vartheta \,; \quad \frac{\partial x_{s3}}{\partial \xi} = -\text{sen}\,\vartheta \,; \quad \frac{\partial x_{s3}}{\partial \eta} = -\text{sen}\,\vartheta$$

$$\frac{\partial y_{s3}}{\partial \vartheta} = (2L-\xi)\,\text{sen}\,\vartheta - \eta\,\text{sen}\,\vartheta \,; \quad \frac{\partial y_{s3}}{\partial \xi} = \cos\vartheta \,; \quad \frac{\partial y_{s3}}{\partial \eta} = \cos\vartheta$$

$$\frac{\partial \Delta l_4}{\partial \vartheta} = -(2L-\xi)\cos\vartheta \,; \quad \frac{\partial \Delta l_4}{\partial \xi} = \text{sen}\,\vartheta \,; \quad \frac{\partial \Delta l_4}{\partial \eta} = 0$$

$$[\Lambda_u] = \begin{bmatrix} 1 & 0 & 0 \\ (L-L_1)\cos\vartheta_0 & -\text{sen}\,\vartheta_0 & 0 \\ (L-L_1)\,\text{sen}\,\vartheta_0 & \cos\vartheta_0 & 0 \\ (2L-L_1-L_2)\cos\vartheta_0 & -\text{sen}\,\vartheta_0 & -\text{sen}\,\vartheta_0 \\ (2L-L_1-L_2)\,\text{sen}\,\vartheta_0 & \cos\vartheta_0 & \cos\vartheta_0 \end{bmatrix}$$

$$[\Lambda_k] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \\ -(2L-L_1)\cos\vartheta_0 & \text{sen}\,\vartheta_0 & 0 \end{bmatrix} \quad [\Lambda_e] = \begin{bmatrix} (2L-L_1-L_2)\cos\vartheta_0 & -\text{sen}\,\vartheta_0 & -\text{sen}\,\vartheta_0 \\ (2L-L_1)\cos\vartheta_0 & -\text{sen}\,\vartheta_0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta l_4}{\partial \vartheta^2} = (2L-\xi)\,\text{sen}\,\vartheta \,; \quad \frac{\partial^2 \Delta l_4}{\partial \vartheta \partial \xi} = \cos\vartheta \,; \quad \frac{\partial^2 \Delta l_4}{\partial \xi^2} = 0$$

$$[k_{el1}] = [k_{el2}] = [k_{el3}] = \emptyset$$
(linear kinematic relationships)

$$[K_{el4}] = k_4\,\Delta l_{04} \begin{bmatrix} (2L-L_1)\,\text{sen}\,\vartheta_0 & \cos\vartheta_0 & 0 \\ \cos\vartheta_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

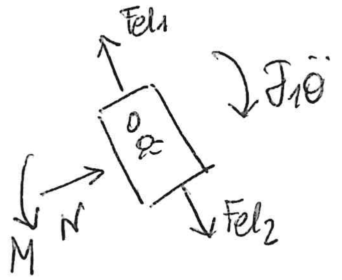
$$h_{s2} = y_{s2} \,; \quad h_{s3} = y_{s3}$$

$$\frac{\partial^2 h_{s2}}{\partial \vartheta^2} = (L-\xi)\cos\vartheta \,; \quad \frac{\partial^2 h_{s2}}{\partial \vartheta \partial \xi} = -\text{sen}\,\vartheta$$

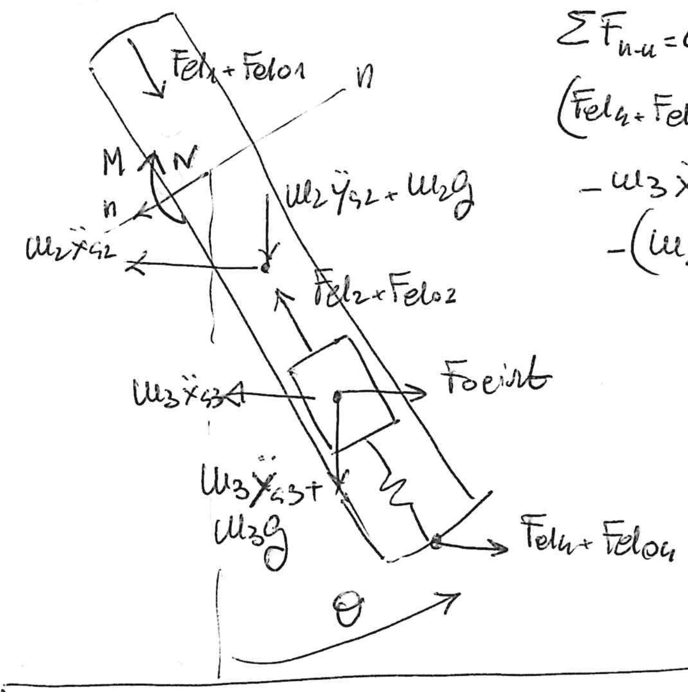
$$\frac{\partial^2 h_{s3}}{\partial \vartheta^2} = (2L-\xi-\eta)\cos\vartheta \,; \quad \frac{\partial^2 h_{s3}}{\partial \vartheta \partial \xi} = -\text{sen}\,\vartheta \,; \quad \frac{\partial^2 h_{s3}}{\partial \vartheta \partial \eta} = -\text{sen}\,\vartheta$$

$$[k_{s2}] = m_2 g \begin{bmatrix} (L-L_1)\cos\vartheta_0 & -\mathrm{sen}\,\vartheta_0 & 0 \\ -\mathrm{sen}\,\vartheta_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [k_{s3}] = m_3 g \begin{bmatrix} (2L-L_1-L_2)\cos\vartheta_0 & -\mathrm{sen}\,\vartheta_0 & -\mathrm{sen}\,\vartheta_0 \\ -\mathrm{sen}\,\vartheta_0 & 0 & 0 \\ -\mathrm{sen}\,\vartheta_0 & 0 & 0 \end{bmatrix}$$

- question #4 : $S_{rel} = M$
- question #6 :



$$\Sigma M_O^{J_1} = 0 \rightarrow \boxed{M = J_1 \ddot{\theta}}$$



$\Sigma F_{n-n} = 0$:

$$(Fel_4 + Felo_4)\cos\vartheta - (m_3 g + m_3 \ddot{y}_{G3})\,\mathrm{sen}\,\vartheta +$$
$$- m_3 \ddot{x}_{G3}\cos\vartheta - m_2 \ddot{x}_{G2}\cos\vartheta +$$
$$-(m_2 g + m_2 \ddot{y}_{G2})\,\mathrm{sen}\,\vartheta + F_0 e^{i\Omega t}\cos\vartheta - N = 0$$

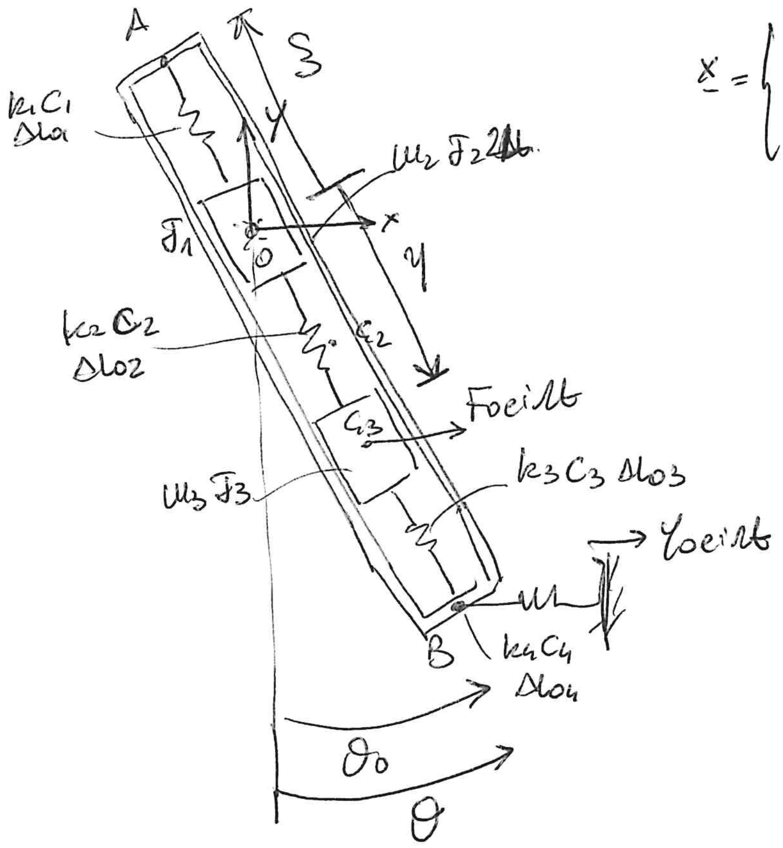
$$\cos\vartheta \simeq \cos\vartheta_0 - \mathrm{sen}\,\vartheta_0\,(\vartheta - \vartheta_0)$$
$$\mathrm{sen}\,\vartheta \simeq \mathrm{sen}\,\vartheta_0 + \cos\vartheta_0\,(\vartheta - \vartheta_0)$$

$$\Downarrow$$

$$\boxed{\begin{aligned} N = {} & \left(Fel_{4_{LIN}} + F_0 e^{i\Omega t} - m_3 \ddot{x}_{G3_{LIN}} - m_2 \ddot{x}_{G2_{LIN}}\right)\cos\vartheta_0 - \left(m_3 \ddot{y}_{G3_{LIN}} + m_2 \ddot{y}_{G2_{LIN}}\right)\mathrm{sen}\,\vartheta_0 + \\ & - \left(Felo_4\,\mathrm{sen}\,\vartheta_0 + m_3 g \cos\vartheta_0 + m_2 g \cos\vartheta_0\right)(\vartheta - \vartheta_0) \end{aligned}}$$

$$Fel_{4_{lin}} = (k_4 + i\Omega c_4)(y - x_{B_{LIN}})$$

2^{nd} set of indepenent coordinates



$$\underline{x} = \begin{Bmatrix} \theta \\ \xi \\ \eta \end{Bmatrix} \qquad \underline{x_0} = \begin{Bmatrix} \theta_0 \\ L_1 \\ 2L - L_1 - L_2 \end{Bmatrix}$$

$$x_{G2} = (L-\xi)\,\mathrm{sen}\,\theta$$
$$y_{G2} = -(L-\xi)\cos\theta$$
$$x_B = (2L-\xi)\,\mathrm{sen}\,\theta$$
$$y_B = -(2L-\xi)\cos\theta$$
$$x_{G3} = \eta\,\mathrm{sen}\,\theta$$
$$y_{G3} = -\eta\cos\theta$$

$$\Delta l_1 = \xi - L_1$$
$$\Delta l_3 = (2L - \xi - \eta) - L_2$$
$$\Delta l_2 = \eta - (2L - L_1 - L_2)$$
$$\Delta l_4 = -\Delta x_B = -(2L-\xi)\,\mathrm{sen}\,\theta$$

$$\frac{\partial x_{G2}}{\partial\theta} = (L-\xi)\cos\theta; \quad \frac{\partial x_{G2}}{\partial\xi} = -\mathrm{sen}\,\theta$$
$$\frac{\partial y_{G2}}{\partial\theta} = (L-\xi)\,\mathrm{sen}\,\theta; \quad \frac{\partial y_{G2}}{\partial\xi} = \cos\theta$$
$$\frac{\partial x_{G3}}{\partial\theta} = \eta\cos\theta; \quad \frac{\partial x_{G3}}{\partial\eta} = \mathrm{sen}\,\theta$$
$$\frac{\partial y_{G3}}{\partial\theta} = \eta\,\mathrm{sen}\,\theta; \quad \frac{\partial y_{G3}}{\partial\eta} = -\cos\theta$$
$$\frac{\partial\Delta l_2}{\partial\theta} = 0; \quad \frac{\partial\Delta l_2}{\partial\xi} = 0; \quad \frac{\partial\Delta l_2}{\partial\eta} = 1$$
$$\frac{\partial\Delta l_3}{\partial\theta} = 0; \quad \frac{\partial\Delta l_3}{\partial\xi} = -1; \quad \frac{\partial\Delta l_3}{\partial\eta} = -1$$

$$[\Lambda_m] = \begin{bmatrix} 1 & 0 & 0 \\ (L-L_1)\cos\theta_0 & -\mathrm{sen}\,\theta_0 & 0 \\ (L-L_1)\,\mathrm{sen}\,\theta_0 & \cos\theta_0 & 0 \\ (2L-L_1-L_2)\cos\theta_0 & 0 & \mathrm{sen}\,\theta_0 \\ (2L-L_1-L_2)\,\mathrm{sen}\,\theta_0 & 0 & -\cos\theta_0 \end{bmatrix}$$

$$[\Lambda_k] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -1 \\ -(2L-L_1)\cos\theta_0 & \mathrm{sen}\,\theta_0 & 0 \end{bmatrix} \qquad [\Lambda_d] = \begin{bmatrix} (2L-L_1-L_2)\cos\theta_0 & 0 & \mathrm{sen}\,\theta_0 \\ (2L-L_1)\cos\theta_0 & -\mathrm{sen}\,\theta_0 & 0 \end{bmatrix}$$

[Kel4] → see previous solution

[Ks2] → " " "

$$\frac{\partial^2 Y_{S3}}{\partial \theta^2} = \eta \cos\theta \qquad \frac{\partial^2 Y_{S3}}{\partial \theta \partial S} = 0 \qquad \frac{\partial^2 Y_{S3}}{\partial \theta \partial \eta} = \operatorname{sen}\theta$$

$$[K_{S3}] = m_3 g \begin{bmatrix} (2L - L_1 - L_2)\cos\theta_0 & 0 & \operatorname{sen}\theta_0 \\ 0 & 0 & 0 \\ \operatorname{sen}\theta_0 & 0 & 0 \end{bmatrix}$$

- question #4 : $S_{rel} = -\xi - \eta$